Who holds the proof?

Housing Tenure, Documentation, and Administrative Burden in Federal Disaster Assistance

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Motivation

Disaster assistance requires documented need, **often after the disaster destroys records**

Eligible claimants **excluded by procedure** (Brodkin and Majmundar 2010; Herd and Moynihan 2018); learning, compliance, psychological costs (Moynihan et al. 2015)

Housing tenure, understudied social vulnerability esp. in disasters (Lee and Van Zandt 2019); One in three U.S. households rent

Proof of occupancy: Renters rely on leases, rent receipts, landlord statements; owners on property records (Aiken et al. 2023)

Gap

Aggregate *approval and award* rates do not show where in the claims process renters fall behind

Research question and contributions

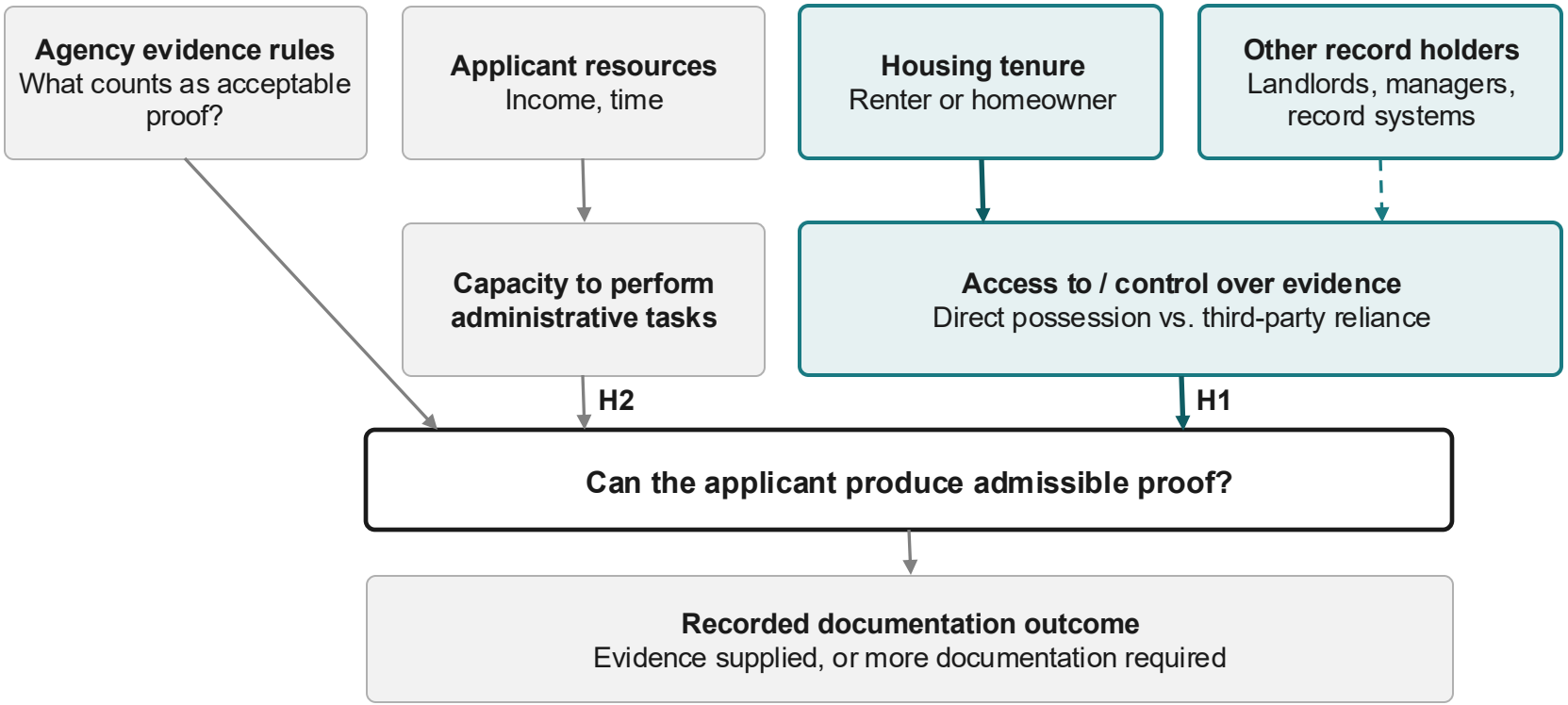
Research question

Do tenure differences reflect who is approved for assistance, or who can supply the evidence needed to establish a claim?

Contributions

- **Theoretical:** *Resources to comply* vs. *Control over evidence*
- **Empirical:** Applicant-level recorded denial grounds, approval and awards; 22.1M FEMA Individuals and Households Program (IHP) applications, 2002–2025

Framework: What determines approval?



Hypotheses

Control over evidence → Documentation

Control by actors outside the citizen–state encounter (Nisar 2018; Barnes 2021; Halling and Baekgaard 2024)

►**H1**: Renters > Homeowners in documentation-related denial

Resources → Documentation

►**H2**: Documentation-related denial declines with income

Procedural knowledge → Approval (Masood and Nisar 2021)

►**H3**: Distance to nearest Disaster Recovery Center (DRC) → Lower approval

►**H4**: County disaster declarations, prior 60 months → Higher approval

Data and design

Records: OpenFEMA IHP valid registrations, Nov. 2002–Oct. 2025 (FEMA 2025)

- 22,095,904 applications; contiguous U.S.; COVID-19 declarations excluded; 57.6% owners, 41.1% renters
- Linked: ZCTA ACS socioeconomic variables (2011–2023), DRC locations, county disaster declarations

Outcomes

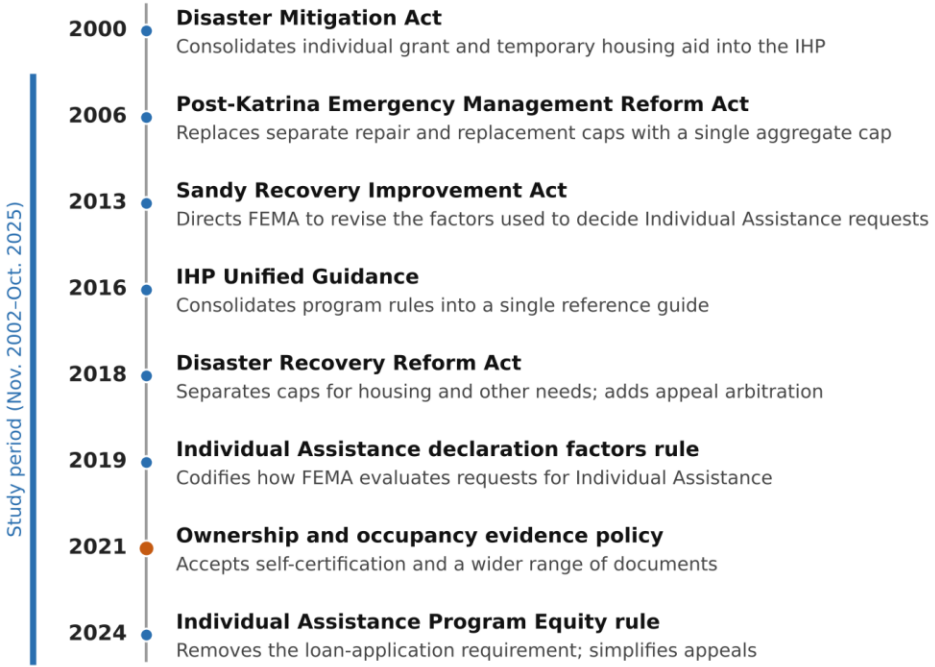
- Approval; award among approved ($, log)

FEMA provides 9 pre-defined categories of denial, non-duplicative

- Documentation-related denial: Additional documentation required
- Substantive denial: No eligible damages & needs, existing insurance

Estimation: Logit, OLS, LPM; ZCTA and year FE; Applicant and ACS controls, flood insurance, verified loss; SEs clustered by ZCTA; year-specific coefficients

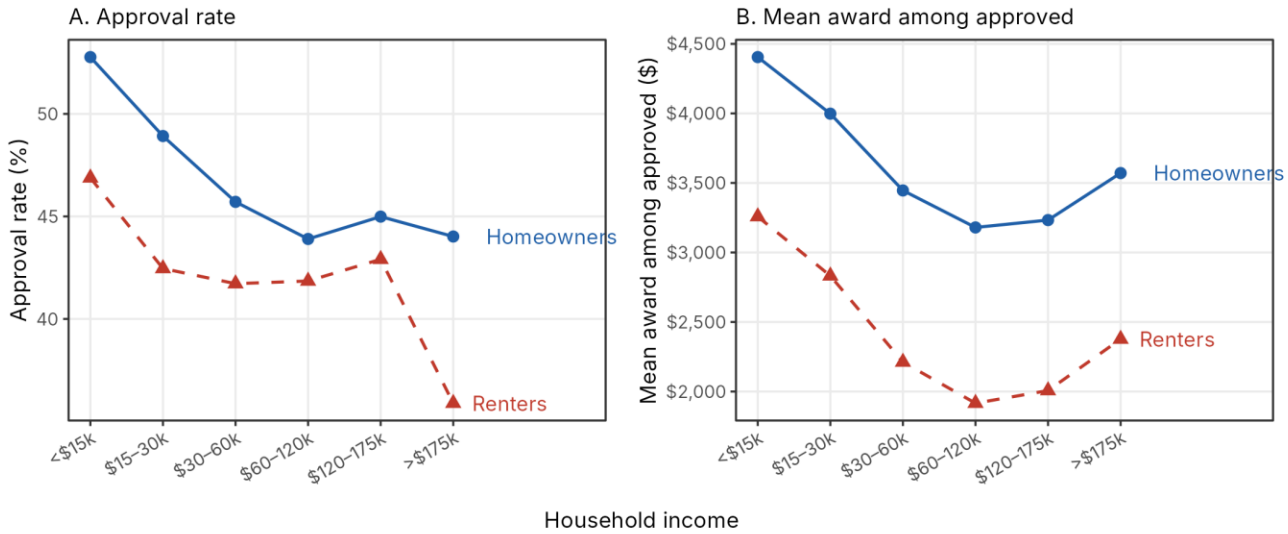
Program context



- Repair/replacement assistance: Owners only
- 2021: Expanded evidence criteria on ownership & occupancy (FEMA 2021)
- Pooled estimates span shifting documentation rules

Descriptive gaps

- ►Approval: 46% owners, 42% renters.
- ►Mean award among approved: $3,725 vs. $2,728.



Aggregate outcomes do not locate the tenure gap

Renter coefficient	Baseline	+ Interactions, flood ins.	+ Repair/replacement elig.	N
Approval (logit)	0.444***	0.396***	0.868***	7,949,403
Award ($)	−900***	−1,664***	1,416***	3,182,809
Award (log)	0.235***	0.103***	0.474***	3,182,809

- ►Adjusted renter approval coefficient positive; reverses descriptive gap
- ►Award contrast changes sign with scale and owner-only eligibility controls
- ►Repair/replacement eligibility: assistance determination, not confounder

Main finding: Renters fall behind on documentation

	Documentation-related	Substantive
Adjusted		
Renter (vs. homeowner)	**2.42*** **	**−16.09*** **
	(0.06)	(0.28)
Adjusted homeowner rate	2.32%	35.79%
Implied renter rate	4.74%	19.69%
N	6,779,452	

LPM, percentage points; SEs clustered by ZCTA.

Across all applications (approved + denied),

- ▶ Documentation-related denial: +2.42 pp, ≈104.5% above homeowner rate
- ▶ Substantive denial: −16.09 pp
- ▶ Opposite signs: different denial grounds
- ▶ Net of submission timing, income, loss, registration channel

H1 supported

Income: No monotonic gradient

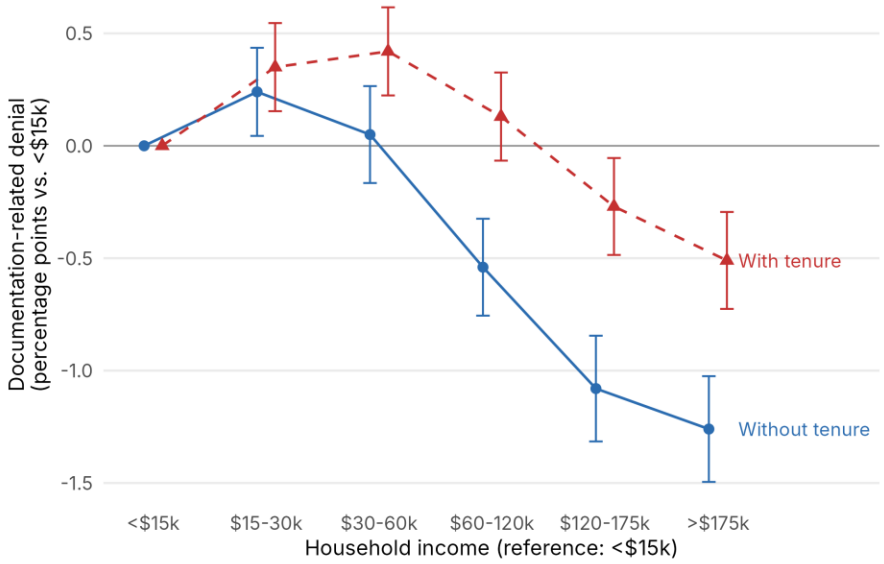


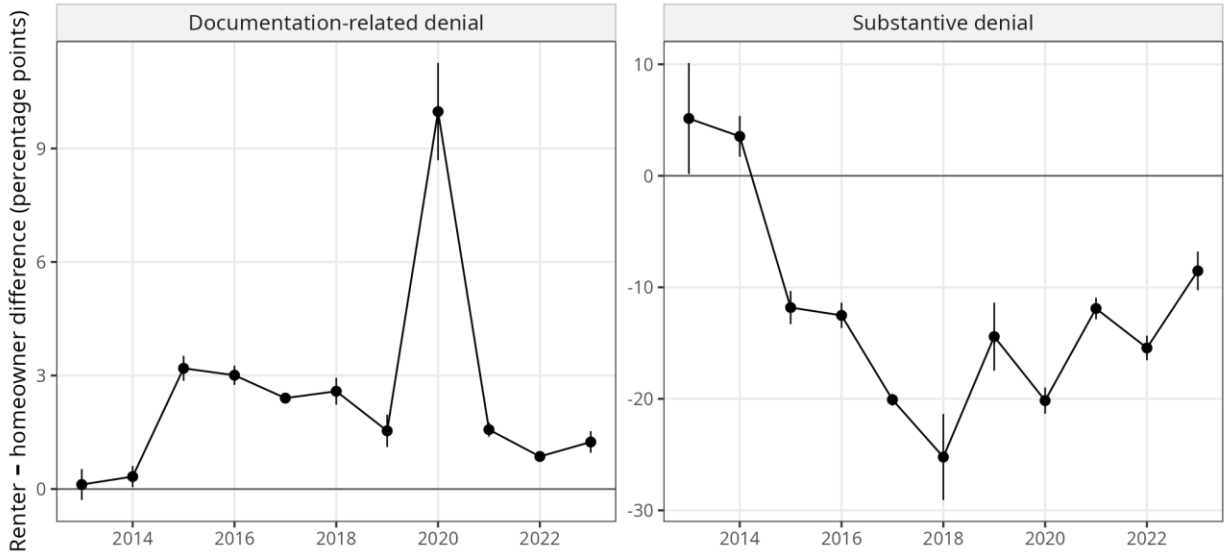
Table 4, columns 1-2; 95% confidence intervals, SEs clustered by ZCTA. N = 6,779,452.

Documentation-related denial, pp

- Advantage only above $120k
- Lower-middle brackets: *higher* denial
- Adding tenure: top-bracket advantage −1.26 → −0.51
- Part of apparent income effect is tenure

H2: mixed at best

Persistence: Renter–homeowner gap by year



Documentation gap > 0, substantive gap < 0, every year 2015–2023.

Peak 2020: 9.97 pp (remote inspections (FEMA 2020)); 2021–2023: 0.86–1.57 pp.

Access and local experience

	Separate				Joint	
	Approval	Log award	Approval	Log award	Approval	Log award
Distance to nearest DRC (mi)	−0.10***	−0.22***			−0.12***	−0.24***
	(0.02)	(0.04)			(0.02)	(0.04)
County declarations, prior 60m			1.94***	0.42*	1.97***	0.40
			(0.16)	(0.21)	(0.14)	(0.21)
N	9,002,720	3,574,454	9,654,932	3,716,701	8,976,515	3,566,067

Approval: LPM, pp. Log award ×100, approved applicants.

- DRC distance p10 → p90 (1.6 → 19.1 mi): approval 40.5% → 38.7% (**H3**); +1 prior declaration: +1.94 pp (**H4**)
- Annual: distance n.s. 6/13 years; declarations < 0 in 2014, 2020; awards inconsistent

Interpretation

Relational basis of compliance

- ► Compliance capacity = resources + control over required evidence
- ► Understanding a request ≠ ability to produce a landlord's record
- ► Income buys follow-up, not authority over another party's documents

Scope condition: Strongest where applicants rely on records held by landlords, employers, organizations; weakest where agencies verify directly (Herd et al. 2013)

Measurement: Pooled approval and awards mask stage-specific disadvantage

Alternative accounts: Renters less visible in public records used for automated checks; higher post-disaster mobility

Implications and limitations

Design

- Agencies can audit documentation requests by who holds the record
- Accepting alternative evidence vs. agency-obtained records with consent
- Assistance through follow-up stages, weeks after disaster and initial application; evaluate resolved requests, appeal reversals

Limitations

- Conditional associations based on observation of administrative records, not causal
- Unobserved: Landlord cooperation, DRC use, experienced costs
- Denial reasons: Administrative classifications; non-applicants excluded

Three takeaways

1. **Finding:** Renters' documentation-related denial ≈2× homeowners'; substantive denial lower; recurs every year 2015–2023

2. **Other factors:** Income advantage confined to top brackets and partly a tenure effect; DRC proximity and county experience raise pooled approval but are unstable annually

3. **Implication:** Burden depends on what agencies require and what applicants bring

Who holds the proof matters.

Thank you

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OLS models of awards

	Dollars			Log		
	(1)	(2)	(3)	(4)	(5)	(6)
Renter (vs. homeowner)	−900***	−1,664***	1,416***	0.235***	0.103***	0.474***
	(63)	(72)	(59)	(0.009)	(0.008)	(0.013)
Flood insurance		−3,953***	−1,768***		−0.654***	−0.372***
		(193)	(133)		(0.020)	(0.012)
Repair assistance eligible			5,797***			0.801***
			(186)			(0.015)
Replacement assistance eligible			16,736***			0.973***
			(1,102)			(0.051)
Verified loss	0.234***	0.241***	0.187***	0.665***	0.681***	0.607***
	(0.018)	(0.019)	(0.018)	(0.006)	(0.005)	(0.006)
Tenure × income interactions	No	Yes	Yes	No	Yes	Yes
Applicant and ACS controls	Yes	Yes	Yes	Yes	Yes	Yes
ZCTA and year FE	Yes	Yes	Yes	Yes	Yes	Yes
N			3,182,809			
R^2	0.393	0.425	0.519	0.599	0.622	0.657

Approved applicants; SEs clustered by ZCTA. With interactions, renter coefficient is the contrast for incomes below $15k. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Denial models

	Documentation-related		Substantive	
	(1)	(2)	(3)	(4)
Renter (vs. homeowner)		2.42***		−16.09***
Ten additional days to submission	0.34***	0.36***	1.27***	1.13***
Registration (vs. call center)				
DSAT	−0.72***	−0.63***	3.93***	3.34***
Internet	−0.50***	−0.46***	−4.33***	−4.60***
Mobile	0.02	−0.15*	−6.16***	−5.00***
Household income (vs. <$15k)				
$15–30k	0.24*	0.35***	0.07	−0.69***
$30–60k	0.05	0.42***	2.26***	−0.18
$60–120k	−0.54***	0.13	6.50***	2.05***
$120–175k	−1.08***	−0.27*	9.41***	4.02***
>$175k	−1.26***	−0.51***	6.93***	1.91***
N		6,779,452		
R^2	0.056	0.059	0.166	0.184

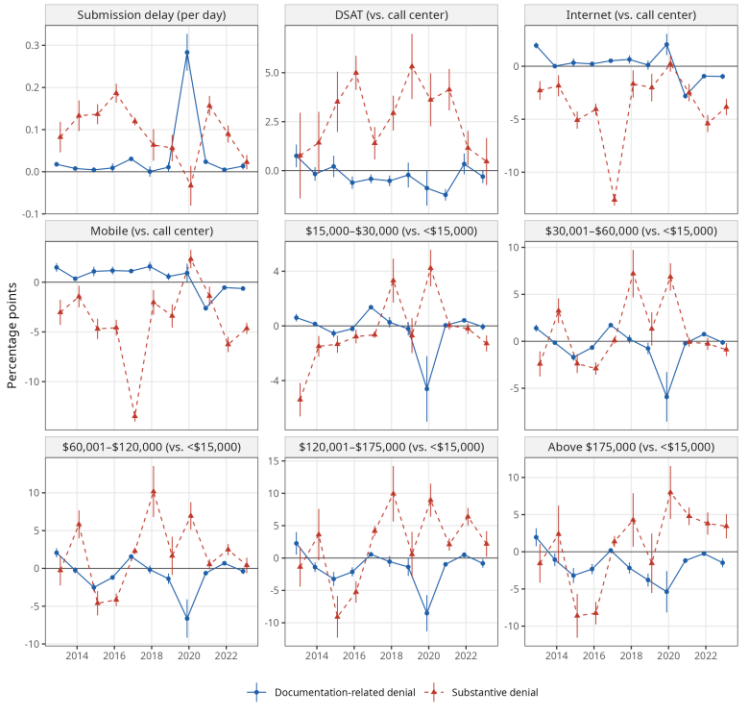
LPM, pp; ZCTA and year FE; FEMA and ACS controls; SEs clustered by ZCTA. *** p < 0.001, ** p < 0.01, * p < 0.05.

Access and experience models

	Distance to DRC		Prior declarations		Joint	
	Approval (1)	Log award (2)	Approval (3)	Log award (4)	Approval (5)	Log award (6)
Distance to nearest DRC (mi)	−0.10***	−0.22***			−0.12***	−0.24***
	(0.02)	(0.04)			(0.02)	(0.04)
County declarations, prior 60 mo.			1.94***	0.42*	1.97***	0.40
			(0.16)	(0.21)	(0.14)	(0.21)
Renter (vs. homeowner)	5.81***	16.23***	6.03***	17.05***	5.83***	16.20***
	(0.21)	(0.82)	(0.20)	(0.78)	(0.21)	(0.82)
Flood insurance	−3.25***	−65.28***	−3.14***	−65.06***	−3.23***	−65.20***
	(0.43)	(2.00)	(0.43)	(1.97)	(0.43)	(2.01)
Log verified loss	10.22***	67.12***	10.22***	67.05***	10.22***	67.12***
	(0.16)	(0.58)	(0.15)	(0.55)	(0.16)	(0.58)
Applicant and ACS controls	Yes	Yes	Yes	Yes	Yes	Yes
ZCTA and year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	9,002,720	3,574,454	9,654,932	3,716,701	8,976,515	3,566,067
R^2	0.319	0.621	0.337	0.618	0.320	0.621

Coefficients ×100; approval: LPM, pp; award: log points ×100, approved applicants. SEs clustered by ZCTA. *** p < 0.001, ** p < 0.01, * p < 0.05.

Annual timing, channel, income associations



Annual DRC distance and prior declarations

